

John Marsden

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EDUCATION	<u>North Carolina State University</u> M.S. – Computer Science Graduation Date: 04/2025
	<u>Abilene Christian University</u> B.S. – Math and Computer Science Graduation Date: 04/2020
SKILLS	<u>Programming Languages:</u> C#, Python, Java, SQL, Rust, HTML, CSS, JavaScript, F#, C++, C, Haskell <u>Databases:</u> Oracle Database, Postgres, MySQL <u>Dev Ops:</u> Fuzzing, Github Actions, Docker, Bash, Linux <u>Frameworks:</u> ASP.NET, Node.js, Rails, Vue, Astro
EXPERIENCE	<u>Teaching Assistant</u> NCSU 07/2020 – 11/2024 - Helped 300+ students navigate the complexities of multi-project Java codebases - Validated 700+ student algorithm analyses and algorithm designs - Graded 400+ student algorithms using common data structures, including stacks, graphs, etc. <u>Instructor - Software Development Fundamentals</u> NCSU 05/2022 – 07/2022 - Taught the software development process and test-driven development to 20 students - Demonstrated idiomatic generic object-oriented Java code to students during 5+ live coding sessions - Taught students a variety of common design patterns and both when and when not to apply them <u>Instructor - Introduction to Computing</u> NCSU 12/2021 – 04/2022 - Taught the fundamentals of Java to 40 students - Built two autograders using JUnit 5 to autograde student code submissions <u>Software Development Intern</u> Secmation 04/2021 – 07/2021 - Planned and implemented 20% of a backend for a programming environment with Python - Outlined an API for the programming environment backend using AsyncAPI - Containerized the programming environment backend using Docker for scalability and ease of use <u>Software Development Intern</u> Faithlife 04/2019 – 07/2019 - Converted the OAuth application management portal from being user specific to a group-based system - Integrated a HaveIBeenPwned check using the k-anonymity API
PROJECTS	<u>General Purpose Discord Bot</u> - Built Norm: A general-purpose Discord bot written with the .NET runtime - Implemented an HTML to Discord Markdown transpiler - Built an event announcement system that supported timezone adjustment and role-based announcements - Created an OMDb integration to allow for users to plan movie nights <u>Natural Language Processing Wrapper Library</u> - Designed a wrapper library around the Microsoft.Recognizers.Text library adding static type support - Created an extensible structure for supporting future recognizers - Created a DateTime recognizer wrapper - Authored a public and private documented library API for ease of future extension and contribution <u>Machine Learning on Intermediate Languages</u> - Created a code embedding machine learning model to convert control-flow graphs (CFG) to vectors - Reimplemented the deepwalk learning algorithm for distributed learning - Constructed an attention network for using CFG embeddings to predict student performance <u>Control Flow Graph Compiler</u> - Implemented a custom intermediate language representation with the necessary semantics for converting programs to control flow graphs (CFG) - Created a compiler in Rust to convert the custom intermediate language to a statement level control flow graph - Implemented a parallelized TCP server on top of the compiler to handle over two hundred thousand compilations necessary for training a machine learning model